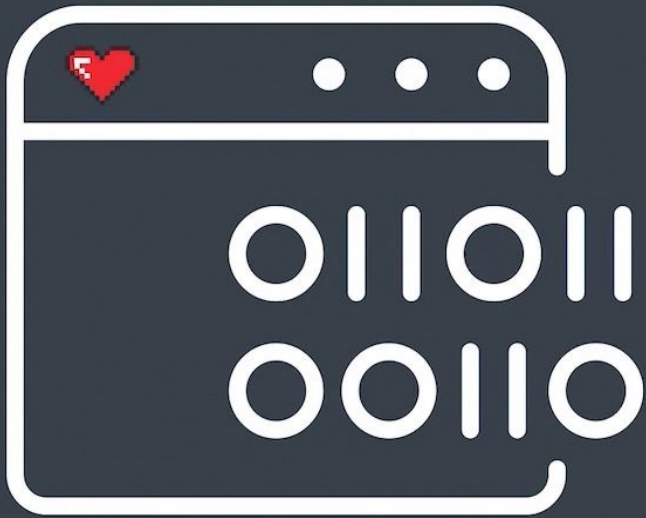


BIT MANIPULATION



Binary Number System

0 & 1

0, 1, 2, 3, 4, 5, 6, 7, 8, 9 $\rightarrow$ 10

decimal
 $\downarrow$
10
 $\rightarrow$

Binary
 $\downarrow$
2
 $\downarrow$ $\downarrow$
0 1

BIT MANIPULATION



Binary Number System 0 & 1

Decimal to Binary (4)

Binary to Decimal (100)

BIT MANIPULATION



Binary Number System 0 & 1

Decimal to Binary (4)

↓
4

2	4	0
2	2	0
	1	

$(100)_2 = (4)_{10}$

$00000100 = 100 = 00100 = 000100$

Binary to Decimal (100)



Binary Number System 0 & 1

Decimal to Binary (4)

00000100

4

2 | 4

0

0

100

00100

000100

$= (4)_{10}$

Binary to Decimal (100)

$1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0$

↓

4

↓

0

↓

0

$(100)_2 = (4)_{10}$

Binary Number System 0 & 1

Decimal to Binary (4)

0 → 000 →

1 → 001

2 → 010

3 → 011

4 100

5 101

6 110

7 111

8 1000



Binary to Decimal (100)



Bit-wise Operators

Binary AND &

Binary OR |

Binary XOR ^

Binary One's Complement ~

Binary Left Shift <<

Binary Right Shift >>

BIT MANIPULATION



Binary AND &

Rules

0	&	0	0
0	&	1	0
1	&	0	0
1	&	1	1

5 & 6

A = 0101

B = 0110

BIT MANIPULATION



BitManipulation.java > BitManipulation > main(String[])

```
1  import java.util.*;
2
3  public class BitManipulation {
4      Run | Debug
5      public static void main(String args[]) {
6          System.out.println((5 & 6));
7      }
```

armankhan082020@gmail.com

PROBLEMS 7 OUTPUT TERMINAL DEBUG CONSOLE

shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java

4

shradhakhapra@Shradhas-Air Classroom Codes %



Binary OR |

Rules

5 | 6

A = 0101

B = 0110

BIT MANIPULATION



Binary OR |

Rules

0		0	0
0		1	1
1		0	1
1		1	1

516

101.
110
1

A = 0101

B = 0110



Binary XOR [^]

Rules

$$5 \wedge 6$$

$$A = 0101$$

$$B = 0110$$

BIT MANIPULATION



Binary One's Complement ~

Rules

$\sim 0 = 1$
 $\sim 1 = 0$

NOT

~ 5
 $\downarrow$
 101
 $\hline$
 $010 \rightarrow$

A = 0101



Binary One's Complement ~

Rules

~ 0 1

~ 1 0

(LSB) Least Significant Bit

(MSB) Most " "

NOT

~5

↓

101

010 → (2)₁₀

A = 0101

5 → 0000101
MSB LSB

① 0000101

↓

-5



Rules

~ 0 1

~ 1 0

~ 5

↓

1 0 1

—

0 1 0 $\rightarrow (2)_{10}$

A = 0101

① 0 0 0 0 1 0 1

↓

-5

{ 1 1 1 1 1 0 1 0 }

↓

$(-6)_{10}$

2's complement

↓

1's complement ($\sim n$)

↓

add 1

Binary One's Complement ~

Rules

~5

A = 0101

$\sim 0 \rightarrow 1$
 $\leftarrow$

$\sim 5 \rightarrow$ 0000101
↓

1111010

↓ 1's comp

0000101

↓ +1

(0000110) $\rightarrow$ (6)₁₀

① 1's comp $\text{not}(n)$
 $\sim(n)$

② add 1



Binary One's Complement ~

Rules

$$\sim 0 \rightarrow 1$$

$$\sim 5 \rightarrow \textcircled{0}0000101$$

$$\textcircled{1}1111010$$

$$\downarrow \text{1's comp}$$
$$00000101$$

$$\downarrow +1$$
$$(00000110) \rightarrow (6)_{10}$$

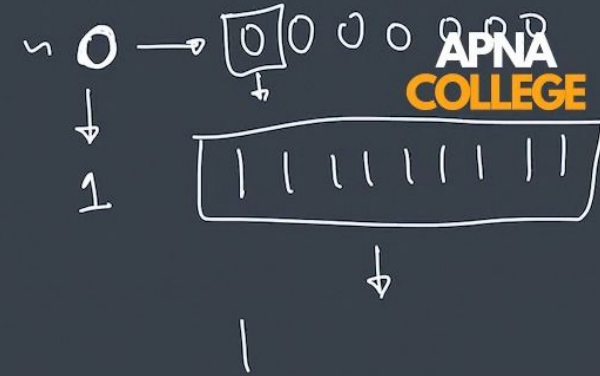
~5

A = 0101

① 1's comp

② add 1

$$\left. \begin{array}{l} \text{not}(n) \\ \sim(n) \end{array} \right\} \rightarrow \text{2's comp}$$
$$\sim 5 = -6$$



Binary Left Shift <<

Rules

$5 \ll 2$

$A = 000101$

BIT MANIPULATION



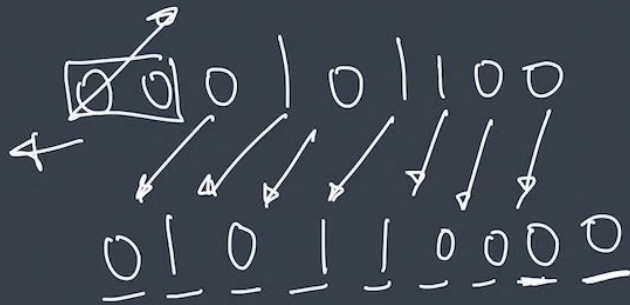
Binary Left Shift <<

Rules

$5 \ll 2$

$A = 000101$

$a \ll b(2)$



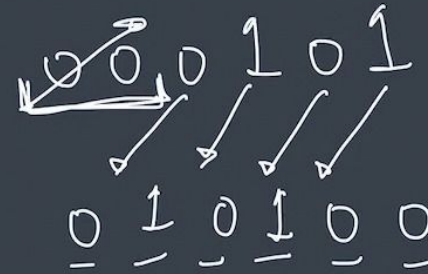
Binary Left Shift <<

Rules

$$a \ll b(2)$$

$$\underline{\underline{5 \ll 2}}$$

$$\underline{\underline{A = 000101}}$$



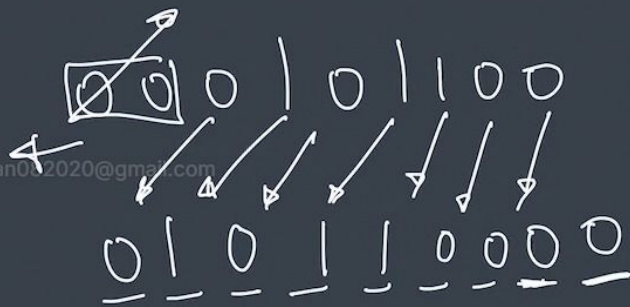
$$5 \ll 2 = (10100)_2 \rightarrow$$



Binary Left Shift <<

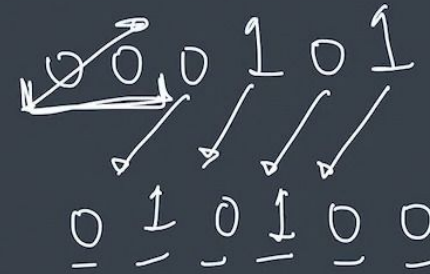
Rules

$$a \ll b(2)$$



$$5 \ll 2$$

$$A = 000101$$



$$5 \ll 2 = (10100)$$

$$\downarrow$$

$$5 \ll 9$$

$$a \ll b = a * 2^b$$



BitManipulation.java > BitManipulation.java

www.apnacollege.in – To exit full screen, press Esc

```
1  import java.util.*;
2
3  public class BitManipulation {
4      public static void main(String args[]) {
5          System.out.println((5<<2));
6      }
7  }
```

PROBLEMS 7 OUTPUT TERMINAL DEBUG CONSOLE

```
3
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
-6
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
-1
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
```

Binary Right Shift >>

Rules

$\xrightarrow{\gg 2}$
 00 | 00 | 00

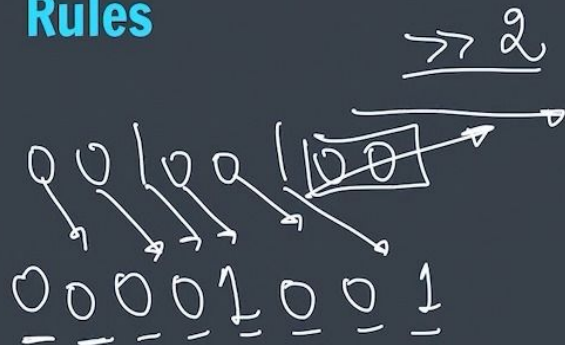
6<<1

A = 000110



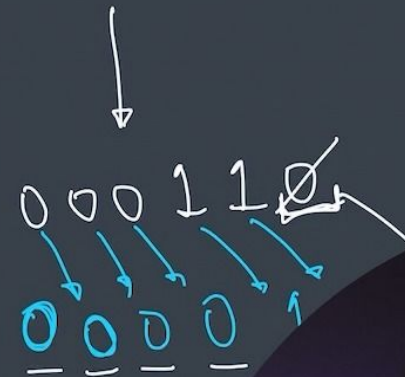
Binary Right Shift

Rules



$$\begin{aligned} \cancel{6} &\ll \cancel{1} \\ 6 &\gg 1 \end{aligned}$$

$$A = \underline{000110}$$



$$6 \gg 1 = (3)_{10}$$

BIT MANIPULATION



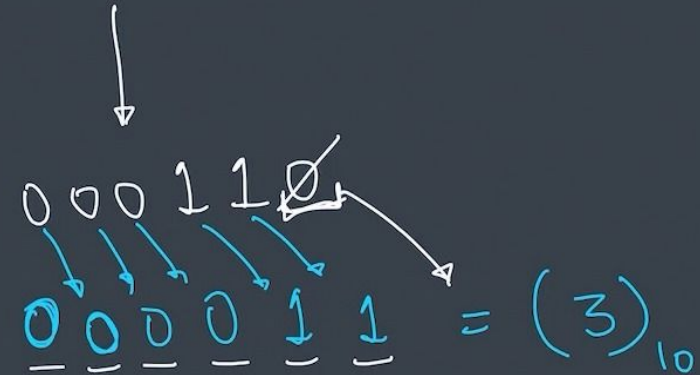
Binary Right Shift >>

Rules



$$\begin{aligned} 6 &\ll 1 \\ 6 &\gg 1 \end{aligned}$$

$$A = 000110$$



$$\gg 1 = (3)_{10}$$

$$a \gg b = a / 2^b$$



Question 1

Check if a number is **Odd or Even**

LSB

[0 = 000
[1 = 001
[2 = 010
[3 = 011
[4 = 100
[5 = 101

odd
↓
LSB →



Question 1

LSB

Check if a number is **Odd or Even**

[0 = 000

[1 = 001

[2 = 010

[3 = 011

[4 = 100

[5 = 101

odd

↓

LSB → 1

even

↓

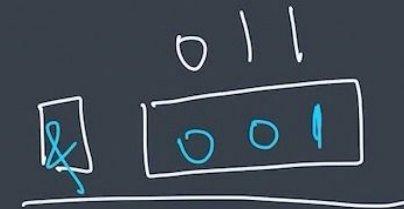
LSB → 0



Question 1

Check if a number is **Odd or Even**

LSB



[0 = 000

[1 = 001

[2 = 010

[3 = 011

[4 = 100

[5 = 101

odd

↓

LSB → 1

even

↓

LSB → 0

$n \& 0 \Rightarrow 0$

$n \& 1$

...

100000001
↓
1

BIT MANIPULATION



armankhan082020@gmail.com

Question 1

Check if a number is **Odd or Even**

[0 = 000]
 [1 = 001]
 [2 = 010]
 [3 = 011]
 [4 = 100]
 [5 = 101]

$$\begin{array}{r}
 3 \\
 \hline
 011 \\
 \& 001 \\
 \hline
 001 \\
 \hline
 = (1)_{10} \\
 \downarrow \\
 \text{odd}
 \end{array}$$

4
1



armankhan082020@gmail.com

Question 1

Check if a number is **Odd or Even**

$$[0 = 000]$$

$$[1 = 001]$$

$$[2 = 010]$$

$$[3 = 011]$$

$$[4 = 100]$$

$$[5 = 101]$$

$$\begin{array}{r} 3 \\ 011 = n \\ \& 001 \\ \hline 001 \\ \hline = (1)_{10} \end{array}$$

↓
odd

$$\begin{array}{r} 4 \\ 100 = n \\ \& 001 \\ \hline 000 \\ \hline = (0)_{10} \end{array}$$

armankhan082020@gmail.com



Question 1

Check if a number is **Odd or Even**

$$[0 = 000]$$

$$[1 = 001]$$

$$[2 = 010]$$

$$[3 = 011]$$

$$[4 = 100]$$

$$[5 = 101]$$

9 → odd

bitmask

$$\begin{array}{r} 1001 \\ \& 0001 \\ \hline 0001 \end{array}$$

$$= (1)_{10} \neq 0$$

↓
odd

$$\begin{array}{r} 3 \\ 011 = n \\ \& 001 \\ \hline 001 \\ \hline = (1)_{10} \end{array}$$

↓
odd

$$\begin{array}{r} 4 \\ 100 = n \\ \& 001 \\ \hline 000 \\ \hline = (0)_{10} \end{array}$$



BitManipulation.java > BitManipulation

```
1  import java.util.*;
2
3  public class BitManipulation {
4      public static void oddOrEven(int n) {
5          int bitMask = 1;
6          if((n & bitMask) == 0) {
7              //even number
8              System.out.println("even number");
9          } else {
10             System.out.println("odd number");
11         }
12     }
13
14     Run | Debug
15     public static void main(String args[]) {
16         oddOrEven(3);
17         oddOrEven(11);
18         oddOrEven(14);
19     }
20 }
```



Operations

1) Get ith bit

0 0 0 1 1 1 1 $i = 2$
 $\Rightarrow$ ith bit = 1

0th bit (LSB) (0 0 0 1) (1) **APNA COLLEGE**
 $\downarrow \downarrow \rightarrow \& 1$
ith bit
 $\downarrow \downarrow \rightarrow \& (1 \ll i)$

Set ith bit

Clear ith bit

BIT MANIPULATION



Operations

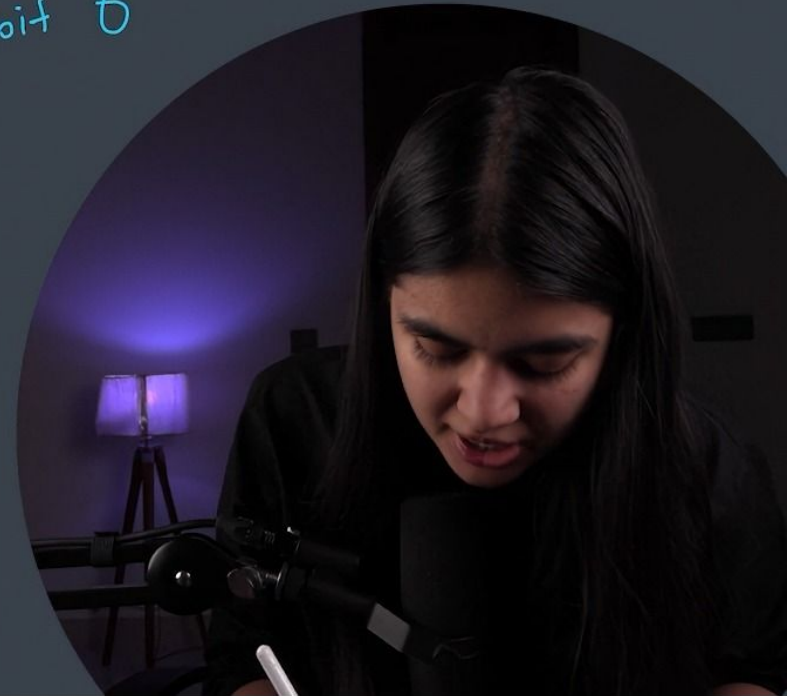
1) Get ith bit

$$\begin{array}{r} \begin{array}{ccccccc} 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ & & & & \textcircled{0} & & \end{array} & i=2 \\ \& [0 & 0 & 0 & 0 & 1 & 0 & 0] \rightarrow 1 \leq i \leq 31 \quad (1 \leq i) \\ \hline 0 & 0 & 0 & 0 & 0 & \textcircled{1} & 0 & 0 \\ \hline & & & & & \downarrow & \\ & & & & & 0 & \rightarrow \text{ith bit } 0 \end{array}$$

$\neq 0 \rightarrow \text{ith bit } 1$

Set ith bit

Clear ith bit



BitManipulation.java > BitManipulation > main(String[])

```
8      System.out.println("even number");
9  } else {
10     System.out.println("odd number");
11 }
12
13
14 public static int getIthBit(int n, int i) {
15     int bitMask = 1<<i;
16     if((n & bitMask) == 0) {
17         return 0;
18     } else {
19         return 1;
20     }
21 }
22
23 Run | Debug
24 public static void main(String args[]) {
25     System.out.println(getIthBit(10, 2));
26 }
27 }
```

armankhan082020@gmail.com



Operations

Get ith bit

2) **Set** ith bit

Clear ith bit

BIT MANIPULATION



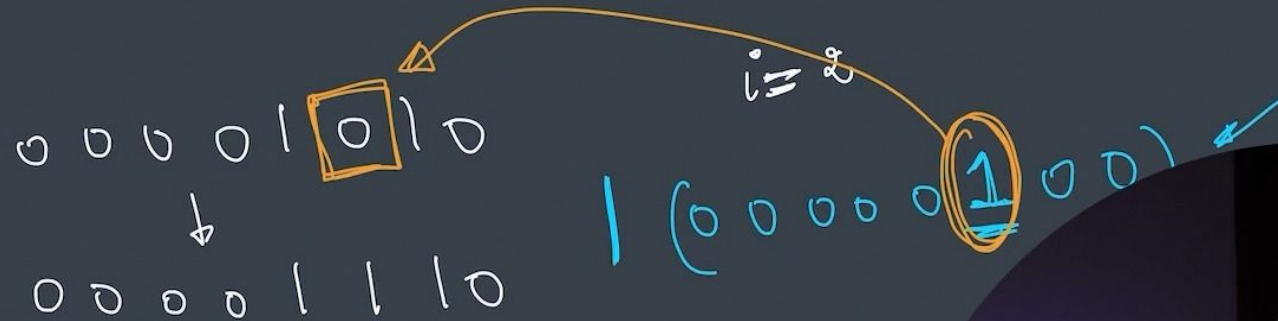
Operations

Get ith bit

number = 1010
n | (1 <= i)

01

2) Set ith bit



Clear ith bit



BitManipulation.java > BitManipulation > main(String[])

```
19         return 1;
20     }
21 }
22
23 public static int setIthBit(int n, int i) {
24     int bitMask = 1<<i;
25     return n | bitMask;
26 }
27
28 public static void main(String args[]) {
29     System.out.println(setIthBit(10, 2));
30 }
```

Run | Debug

PROBLEMS 7 OUTPUT TERMINAL DEBUG CONSOLE

```
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
0
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
1
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
0
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
14
shradhakhapra@Shradhas-Air Classroom Codes %
```

Operations

Get ith bit

2) Set ith bit

Clear ith bit

BIT MANIPULATION

www.apnacollege.in - To exit full screen, press Esc

$$\begin{array}{r}
 10100 \\
 \hline
 1110
 \end{array}
 = 8 + 4 + 2 = 14$$

number = 1010

$n \mid (1 \ll i)$

$$\begin{array}{r}
 0100 \\
 \hline
 1010 \\
 \hline
 \text{set}
 \end{array}$$

14

$i = 2$

$$\begin{array}{r}
 00001010 \\
 \downarrow \\
 00001110
 \end{array}$$

(00000100)



Operations

Get ith bit

Set ith bit

3) Clear ith bit

BIT MANIPULATION

10 = 1010
1000 (8)

i = 1

$1 < i$

↓

$\sim(1 < i)$

4

1010
1101

1000 = 8

$\sim(1 < i) \Rightarrow \sim(0010)$
= 1101



BitManipulation.java > BitManipulation > clearIthBit(int, int)

```
24     int bitMask = 1<<i;
25     return n | bitMask;
26 }
27
28 public static int clearIthBit(int n, int i) {
29     int bitMask = ~(1<<i);
30     return n & bitMask;
31 }
32
33 Run | Debug
34 public static void main(String args[]) {
35     System.out.println(setIthBit(10, 2));
36 }
37 }
```

PROBLEMS 7 OUTPUT TERMINAL DEBUG CONSOLE

```
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
0
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
1
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
0
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
14
shradhakhapra@Shradhas-Air Classroom Codes %
```



BitManipulation.java > BitManipulation.java

www.apnacollege.in - To exit full screen, press Esc

```
24     int bitMask = 1<<i>,
25     return n | bitMask;
26 }
27
28 public static int clearIthBit(int n, int i) {
29     int bitMask = ~(1<<i);
30     return n & bitMask;
31 }
32
33 Run | Debug
34 public static void main(String args[]) {
35     System.out.println(clearIthBit(10, 1));
36 }
37 }
```

PROBLEMS 7 OUTPUT TERMINAL DEBUG CONSOLE

```
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
1
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
0
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
14
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
8
shradhakhapra@Shradhas-Air Classroom Codes %
```

Update ith bit $\xrightarrow{\quad} 0$ + ith
 $\xrightarrow{\quad} 1$

val = 0 or 1

i + (clear, set)



BitManipulation.java > BitManipulation > updateIthBit(int, int, int)

```

30  return n & ~mask;
31  }
32
33  public static int updateIthBit(int n, int i, int newBit) {
34      // if(newBit == 0) {
35      //     return clearIthBit(n, i);
36      // } else {
37      //     return setIthBit(n, i);
38      // }
39
40
41  }
42
43  public static void main(String args[]) {
44      System.out.println(clearIthBit(10, 1));

```

Run | Debug

PROBLEMS 8 OUTPUT TERMINAL DEBUG CONSOLE

```

shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
1
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
0
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
14
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
8
shradhakhapra@Shradhas-Air Classroom Codes %

```



Update ith bit

val = 0 or 1

i + (clear, set)

clear

nB = 0



$nB \ll i$

$0 \ll i (0) \mid n$

$\Rightarrow n$

set nB = 1



$(nB \ll i) \mid n$

↓
ith bit set



Update ith bit

0
+ ith
1

val = 0 or 1

i + (clear, set)

✓ clear

nb = 0

↓
 $nb \ll i$
 $0 \ll i$ (0) | n
↓ cleared
⇒ n

set nb = 1

↓
 $(nb \ll i)$ | n
↓ cleared
ith bit set



Update ith bit

0
+ ith
1

val = 0 or 1

i + (clear, set)

✓ clear

nB = 0

↓
 $nB \ll i$
 $0 \ll i$ (0) | n
↓ cleared
⇒ n
no change

set nB = 1

↓
 $(nB \ll i)$ | n
↓ cleared
↓
ith bit set



BitManipulation.java > BitManipulation > main(String[])

```
34     // if(newBit == 0) {
35     //     return clearIthBit(n, i);
36     // } else {
37     //     return setIthBit(n, i);
38     // }
39
40     n = clearIthBit(n, i);
41     int BitMask = newBit<<i;
42     return n | BitMask;
43 }
44
45 Run | Debug
46 public static void main(String args[]) {
47     System.out.println(clearIthBit(10, 1));
48 }
49 }
```

PROBLEMS 7 OUTPUT TERMINAL DEBUG CONSOLE

```
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
1
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
0
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
14
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
8
shradhakhapra@Shradhas-Air Classroom Codes %
```



BitManipulation.java > BitManipulation > updateIthBit(int, int, int)

```
34 // if(newBit == 0) {
35 //     return clearIthBit(n, i);
36 // } else {
37 //     return setIthBit(n, i);
38 // }
39
40 n = clearIthBit(n, i);
41 int BitMask = newBit<<i;
42 return n | BitMask;
43 }
44
45 public static void main(String args[]) {
46     System.out.println(clearIthBit(10, 1));
47 }
```

Run | Debug

armankhan082120@gmail.com

PROBLEMS 7 OUTPUT TERMINAL DEBUG CONSOLE

```
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
1
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
0
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
14
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
8
shradhakhapra@Shradhas-Air Classroom Codes %
```

BitManipulation.java > BitManipulation > main(String[])

```
34     // if(newBit == 0) {
35     //     return clearIthBit(n, i);
36     // } else {
37     //     return setIthBit(n, i);
38     // }
39
40     n = clearIthBit(n, i);
41     int BitMask = newBit<<i;
42     return n | BitMask;
43 }
44
45 public static void main(String args[]) {
46     System.out.println(updateIthBit(10, 2, 1));
47 }
```

Run | Debug

arman.khan082020@gmail.com

PROBLEMS 7 OUTPUT TERMINAL DEBUG CONSOLE

```
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
1
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
0
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
14
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
8
shradhakhapra@Shradhas-Air Classroom Codes %
```



BitManipulation.java > BitManipulation > main(String[])

```
31     }
32
33     public static int updateIthBit(int n, int i, int newBit) {
34         // if(newBit == 0) {
35         //     return clearIthBit(n, i);
36         // } else {
37         //     return setIthBit(n, i);
38         // }
39
40         n = clearIthBit(n, i);
41         int BitMask = newBit<<i;
42         return n | BitMask;
43     }
44
45     Run | Debug
46     public static void main(String args[]) {
```

armankhan082020@gmail.com

PROBLEMS 7 OUTPUT TERMINAL DEBUG CONSOLE

```
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
0
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
14
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
8
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
14
shradhakhapra@Shradhas-Air Classroom Codes %
```



Clear Last i bits

(15) n=1111, i=2

↓

1 1 0 0

 | |

 ↓ ↓

 i-2 i-1

BIT MANIPULATION



↓

1 1 0 0

↓ ↓

121 120

& ||||| 00

$(i) < i \rightarrow$

00001
 $\leftarrow$

000
 $\leftarrow$
 $i-1$

(~ 0)
 (-1)

$|||||$

 $(\quad)$

Clear Last i bits

(15) n=1111, i=2

↓
1100

& 111100

n
(-1) << i

3 2 1 0
1 1 1 1
& 1100
1100



(1) << i
(~0)
(-1)

11111111 (-1) = ~0
← i

111110000
0

(-1) << i
or
~0 << i

BitManipulation.java > BitManipulation > main(String[])

```
38     // }
39
40     n = clearIthBit(n, i);
41     int BitMask = newBit<<i;
42     return n | BitMask;
43 }
44
45 public static int clearIBits(int n, int i) {
46     int bitMask = (~0)<<i;
47     return n & bitMask;
48 }
49
50 public static void main(String args[]) {
51     System.out.println(clearIBits(15, 2));
52 }
```

Run | Debug

PROBLEMS

7

OUTPUT

TERMINAL

DEBUG CONSOLE

Filter (e.g. text, **/)

ArraysCC.java 1

⚠ The import java.util is never used Java(268435844) [1, 8]

BasicSorting.java 2

⚠ The import java.util.Arrays is never used Java(268435844) [1, 8]

⚠ The import java.util.Collections is never used Java(268435844) [2, 8]

BitManipulation.java 1



Clear Last i bits

(15) n=1111, i=2

↓
1100

& 111100

n
(-1) < i

3 2 1 0
|||
|||

& 1100

1100

$$1100 = 2^3 + 2^2 = 8 + 4 = 12$$

(1) < i

(~0)

(-1)

00001
000
← i-1

11111111 (-1) = ~0
← i

111110000
0

(-1) < i
or
~0 < i



Clear Range of bits

$n=100111010011$, $i=2$, $j=7$

$i=2, j=7$

1 0 0 1 1 1 0 1 0 0 1 1
 j=7 i=2

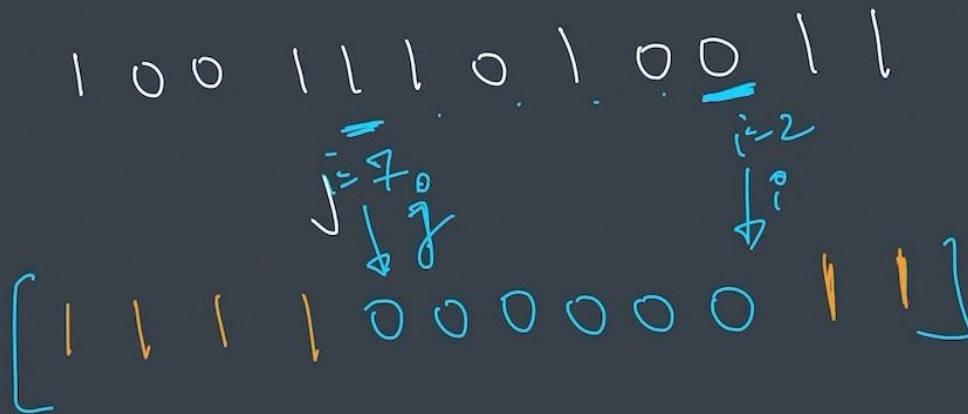
1 0 0 1 0 0 0 0 0 0 1 1



Clear Range of bits

$n=100111010011$, $i=2$, $j=7$

$i=2, j=7$



Resulting bit string: 1001000011



Clear Range of bits

$n=100111010011$, $i=2$, $j=7$

$$\rightarrow a = (n0) \ll (j+1)$$

$$i=2, j=7$$

$a \mid b$

$$\begin{aligned} a &= 1111000000000000 \\ b &= 0000000000000011 \end{aligned}$$

$$01 = (1)_{10} = 2^1 - 1$$

$$011 = (3)_{10} = 2^2 - 1$$

$$0111 = (7)_{10} = 2^3 - 1$$

$$01111 = (15)_{10} = 2^4 - 1$$

$$011111 = (31)_{10} = 2^5 - 1$$



Clear Range of bits

$n=100111010011$, $i=2$, $j=7$

$a|b$

$$\rightarrow \underline{a} = (\sim 0) \ll (j+1)$$

$$\underline{b} = (1 \ll i) - 1$$

$$\downarrow i$$

$$2^i - 1$$

$$i=2, j=7$$

$a|b$

$$\underline{a} = 1111000000000000$$

$$\underline{b} = 0000000000000011$$

$$01 = (1)_{10} = 2^1 - 1$$

$$011 = (3)_{10} = 2^2 - 1$$

$$0111 = (7)_{10} = 2^3 - 1$$

$$01111 = (15)_{10} = 2^4 - 1$$

$$011111 = (31)_{10} = 2^5 - 1$$

$$2^b \rightarrow 1 \ll b$$

$$= 1 \times (2^b)$$



BitManipulation.java > BitManipulation > clearIBits(int, int, int)

```
38     // }
39
40     n = clearIthBit(n, i);
41     int BitMask = newBit<<i;
42     return n | BitMask;
43 }
44
45 public static int clearIBits(int n, int i) {
46     int bitMask = (~0)<<i;
47     return n & bitMask;
48 }
49
50 public static int clearIBits(int n, int i, int j) {
51     int a = ((~0)<<(j+1));
52     int b = (1<<i)-1;
53     int bitMask = a | b;
54     return n & bitMask;
55 }
56
57 Run | Debug
58 public static void main(String args[]) {
59     System.out.println(clearIBits(15, 2));
60 }
```



BitManipulation.java > BitManipulation > main(String[])

```
38     // }
39
40     n = clearIthBit(n, i);
41     int BitMask = newBit<<i;
42     return n | BitMask;
43 }
44
45 public static int clearIBits(int n, int i) {
46     int bitMask = (~0)<<i;
47     return n & bitMask;
48 }
49
50 public static int clearBitsinRange(int n, int i, int j) {
51     int a = ((~0)<<(j+1));
52     int b = (1<<i)-1;
```

PROBLEMS 7 OUTPUT TERMINAL DEBUG CONSOLE

```
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.ja
2
shradhakhapra@Shradhas-Air Classroom Codes %
```



Question 2

Check if a number is a **Power of 2 or not.**

$$\left. \begin{array}{l} 4 \rightarrow 2^2 \\ 8 \rightarrow 2^3 \\ 7 \rightarrow 2^n \times \end{array} \right\}$$



Question 2

Check if a number is a **Power of 2 or not.**

$$\left. \begin{array}{l} 4 \rightarrow 2^2 \\ 8 \rightarrow 2^3 \\ 7 \rightarrow 2^n \times \end{array} \right\} \rightarrow \begin{array}{l} 4 \rightarrow 100 \quad \& \rightarrow \underline{\underline{0}} \\ 3 \rightarrow 011 \\ \hline 8 \rightarrow 1000 \quad \& \rightarrow 0 \\ 7 \rightarrow 0111 \\ \hline 16 \rightarrow 10000 \quad \& \rightarrow 0 \\ 15 \rightarrow 01111 \end{array}$$



Question 2

Check if a number is a **Power of 2 or not.**

$$\begin{array}{l}
 4 \rightarrow 2^2 \\
 8 \rightarrow 2^3 \\
 7 \rightarrow 2^n \times
 \end{array}
 \left. \vphantom{\begin{array}{l} 4 \\ 8 \\ 7 \end{array}} \right\} \rightarrow
 \begin{array}{l}
 4 \rightarrow 100 \quad \& \rightarrow \underline{0} \\
 3 \rightarrow 011 \\
 \hline
 8 \rightarrow 1000 \quad \& \rightarrow 0 \\
 7 \rightarrow 0111 \\
 \hline
 16 \rightarrow 10000 \quad \& \rightarrow 0 \\
 15 \rightarrow 01111
 \end{array}$$

$$\underline{n} = 2^x$$

$$n \& (n-1) = 0$$

$$\begin{array}{l}
 16 \rightarrow 100000 \\
 15 \rightarrow 011111 \quad \& \Rightarrow 0
 \end{array}$$



BitManipulation.java > BitManipulation > isPowerOfTwo(int)

```
49
50     public static int clearBitsinRange(int n, int i, int j) {
51         int a = ((~0)<<(j+1));
52         int b = (1<<i)-1;
53         int bitMask = a | b;
54         return n & bitMask;
55     }
56
57     public static boolean isPowerOfTwo(int n) {
58         return (n&(n-1)) == 0;
59     }
60
61     public static void main(String args[]) {
62         System.out.println(clearBitsinRange(10, 2, 4));
63     }
```

Run | Debug



BitManipulation.java > BitManipulation > main(String[])

```
55     }
56
57     public static boolean isPowerOfTwo(int n) {
58         return (n & (n - 1)) == 0;
59     }
    Run | Debug
60     public static void main(String args[]) {
61         System.out.println(isPowerOfTwo(1));
62     }
63 }
```

PROBLEMS 7 OUTPUT TERMINAL DEBUG CONSOLE

```
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
true
shradhakhapra@Shradhas-Air Classroom Codes %
```



Question 3

Count **Set** Bits in a Number.

|

BIT MANIPULATION



Question 3

Count **Set** Bits in a Number.

10 $\rightarrow$ 1010

no. of set bits = 2

odd & even $\rightarrow$ $\leftarrow (1 \ll 0)$

(i) $n = 1010$
LSB

count = 0

$n \gg 1$

(ii) 0101
LSB

count = 1

$n \gg 1$

(iii) 0010
LSB

count = 1

$n \gg 1$

(iv) 0001
LSB

count = 2



Question 3

Count **Set** Bits in a Number.

10 $\rightarrow$ 1010

no. of set bits = 2

odd & even $\rightarrow$ $\< (1 \ll 0)$

(i) $n = 1010$
LSB

count = 0

$n \gg 1$

(ii) 0101
LSB

count = 1

$n \gg 1$

(iii) 0010
LSB

count = 1

$n \gg 1$

(iv) 0001
LSB

count = 2

(v) 0000



Question 3

Count **Set** Bits in a Number.

10 $\rightarrow$ 1010
no. of set bits = 2

odd & even $\rightarrow$ $\leftarrow (1 \ll 0)$

(i) $n = 1010$
LSB
 $n \gg 1$

count = 0

(ii) 0101
LSB
 $n \gg 1$

count = 1

(iii) 0010
LSB
 $n \gg 1$

count = 1

(iv) 0001
LSB

count = 2

(v) 0000

$\downarrow$
count = 2



Question 3

Count **Set** Bits in a Number.

10 → 1010
no. of set bits = 2

$\log n$

odd & even → $(1 \ll 0)$

(i) $n = 1010$
 $n \gg 1$
LSB

(ii) 0101
 $n \gg 1$
LSB

(iii) 0010
 $n \gg 1$
LSB

(iv) 0001
LSB

(v) 0000
↓
count = 2

count = 0

count = 1

count = 1

co

$\log n$



Question 3

Count **Set** Bits in a Number.

10 → 1010

no. of set bits = 2

$n \rightarrow \log_2 n + 1$
15 → 01111

16 → 10000
 $= \log_2 16 + 1 = 4 + 1 = 5$

odd & even → $(1 \ll 0)$

(i) $n = 1010$
 $n \gg 1$
LSB

(ii) 0101
 $n \gg 1$
LSB

(iii) 0010
 $n \gg 1$
LSB

(iv) 0001
LSB

(v) 0000

↓
count = 2

count = 0

count = 1

count = 1

co

$\log_2 n$



Question 3

Count **Set** Bits in a Number.

10 $\rightarrow$ 1010

no. of set bits = 2

$n \rightarrow \log_2 n + 1$ $O(\log n)$

15 $\rightarrow$ 01111

16 $\rightarrow$ 10000

$$= \log_2 16 + 1 = 4 + 1 = 5$$

odd & even $\rightarrow$ $(1 \ll 0)$

(i) $n = 1010$
LSB
 $n \gg 1$

(ii) 0101
LSB
 $n \gg 1$

(iii) 0010
LSB
 $n \gg 1$

(iv) 0001
LSB

(v) 0000

$\downarrow$
count = 2

count = 0

count = 1

count = 1

co

$\log n$



```
BitManipulation.java > BitManipulation > main(String[])
63     while(n > 0) {
64         if((n & 1) != 0) { //check our LSB
65             count++;
66         }
67         n = n>>1;
68     }
69
70     return count;
71 }
Run | Debug
72 public static void main(String args[]) {
73     System.out.println(isPowerOfTwo(15));
74 }
75 }
```

PROBLEMS 7 OUTPUT TERMINAL DEBUG CONSOLE

```
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
true
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
false
shradhakhapra@Shradhas-Air Classroom Codes %
```



```
BitManipulation.java > BitManipulation > main(String[])  
63     while(n > 0) {  
64         if((n & 1) != 0) { //check our LSB  
65             count++;  
66         }  
67         n = n>>1;  
68     }  
69  
70     return count;  
71 }  
Run | Debug  
72 public static void main(String args[]) {  
73     System.out.println(countSetBits(15));  
74 }  
75 }
```

PROBLEMS 7 OUTPUT TERMINAL DEBUG CONSOLE

```
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java  
true  
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java  
false  
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java  
2  
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java  
4  
shradhakhapra@Shradhas-Air Classroom Codes %
```



Fast Exponentiation

$a^N \rightarrow$ binary form

3^5

5^3

$$\begin{array}{c} a^n \rightarrow \underbrace{a \cdot a \cdot a \cdot a}_{O(n)} \rightarrow n \\ \downarrow \\ O(\log n) \end{array}$$



Fast Exponentiation

$$a^N$$

$$3^5$$

$$5^3$$

$$3^{101}$$

$$\begin{array}{ccc} a^4 & a^2 & a \\ \downarrow & \downarrow & \downarrow \\ 10 & 1 & \end{array}$$

$$a = a^5$$

$$(1 \times a^4) * (1) * (1 \times a) = a^4 \times a^1 = a^5$$

armankhan082020@gmail.com



Fast Exponentiation

a^N

2^5



3^{101}

①

$$\underline{a=3}; \text{ans}=1$$

$$\text{ans}=3$$

$$a = a^2 = \underline{9}$$

②

$$a=9; \text{ans}=3$$

$$\text{ans}=3 \times 1$$

$$a = a^2 = \underline{81} (a^4)$$

③

$$a=81; \text{ans}=3$$

$$\text{ans}=3 \times 81 = 243$$

$$a = a^2 = (81)^2$$

$$\begin{array}{c} (a^4) \quad | \quad 0 \quad | \\ 3 \quad (a^2) \quad (a^1) \end{array}$$

Fast Exponentiation

a^N

2^5



3^{101}

①

$$\underline{a=3}; \text{ans}=1$$

$$\text{ans}=3$$

$$a=a^2=\underline{9}$$

②

$$a=9; \text{ans}=3$$

$$\text{ans}=3 \times 1$$

$$a=a^2=\underline{81} (a^4)$$

③

$$a=81; \text{ans}=3$$

$$\text{ans}=3 \times 81 = 243$$

$$a=a^2=(81)^2$$

$$\begin{array}{c} (a^4) \\ \boxed{1 \ 0 \ 1} \\ 3 \quad (a^2) \quad (a^1) \end{array}$$

$$\underline{\underline{\text{ans} = 243}}$$

5
3 **APNA COLLEGE**

$$\log n + 1$$
$$2+1 \Rightarrow \underline{3}$$

BitManipulation.java > BitManipulation > fastExpo(int, int)

```
70     return count;
71 }
72
73 public static int fastExpo(int a, int n) {
74     int ans = 1;
75
76     while(n > 0) {
77         if((n & 1) != 0) { //check LSB
78             ans = ans * a;
79         }
80         a = a * a;
81         n = n >> 1;
82     }
83 }
```

armankhan082020@gmail.com

PROBLEMS 8 OUTPUT TERMINAL DEBUG CONSOLE

```
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
false
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
2
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
4
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
1
shradhakhapra@Shradhas-Air Classroom Codes %
```



BitManipulation.java > BitManipulation > fastExpo(int, int)

```
71     }
72
73     public static int fastExpo(int a, int n) {
74         int ans = 1;
75
76         while(n > 0) {
77             if((n & 1) != 0) { //check LSB
78                 ans = ans * a;
79             }
80             a = a * a;
81             n = n >> 1;
82         }
83
84         return ans;
85     }
```

armankhan082020@gmail.com

PROBLEMS 7 OUTPUT TERMINAL DEBUG CONSOLE

```
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
false
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
2
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
4
shradhakhapra@Shradhas-Air Classroom Codes % java BitManipulation.java
1
shradhakhapra@Shradhas-Air Classroom Codes %
```



Fast Exponentiation

$$a^N$$

$$5^3 \rightarrow$$

$$\begin{array}{ccc} 5^4 & 5^2 & 5^1 \\ \uparrow & \uparrow & \uparrow \\ 0 & 1 & 1 \end{array}$$

①

ans = 1

$$\underline{3^5}$$

armankhan082@gmail.com

$$\underline{5^3}$$

BIT MANIPULATION



5.5.5

②

$$\text{ans} = 5 \times \underline{25} = 125$$

$$a = (25)^2 = 625$$

$$\underline{n = 000}$$

③

Fast Exponentiation

a^N

2^5

$$5^3 \rightarrow 5$$



5.5.5 → 3 times

||

ans = 1 ; a = 5 ; n = 011

APNA
COLLEGE

①

ans = $1 \times 5 = 5$

a = 25

n = 001

②

ans = $5 \times 25 = 125$

a = $(25)^2 = 625$

n = 000

③



Fast Exponentiation

$$a^N$$

$$3^5$$

$$5^3$$



$$5^3 \rightarrow 5$$

5, 5, 5 $\rightarrow$ 3 times

$$\frac{5, 5, 5}{1,000,000} \rightarrow (\log_2 10^5 + 1)$$

5 $\downarrow$

5 x 2... + 1
constant

$$ans = 1 ; a = 5 ; n = 011$$

①

$$ans = 1 \times 5 = 5$$

$$a = 25$$

$$n = 00\underline{1}$$

②

$$ans = 5 \times 25 = 125$$

$$a = (25)^2 = 625$$

$$n =$$

③



Fast Exponentiation

modular exponentiation
↓

$$a^N$$

$$\underline{3^5}$$

$$5^3$$

$$a^n$$
$$\underline{\underline{a^n \% x}}$$



Fast Exponentiation

$$a^N$$

$$\underline{3^5}$$

$$5^3$$

modular exponentiation

↓
Google

$$\begin{array}{c} a^n \\ \underline{\underline{a^n \% x}} \end{array}$$

